



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI STAR AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

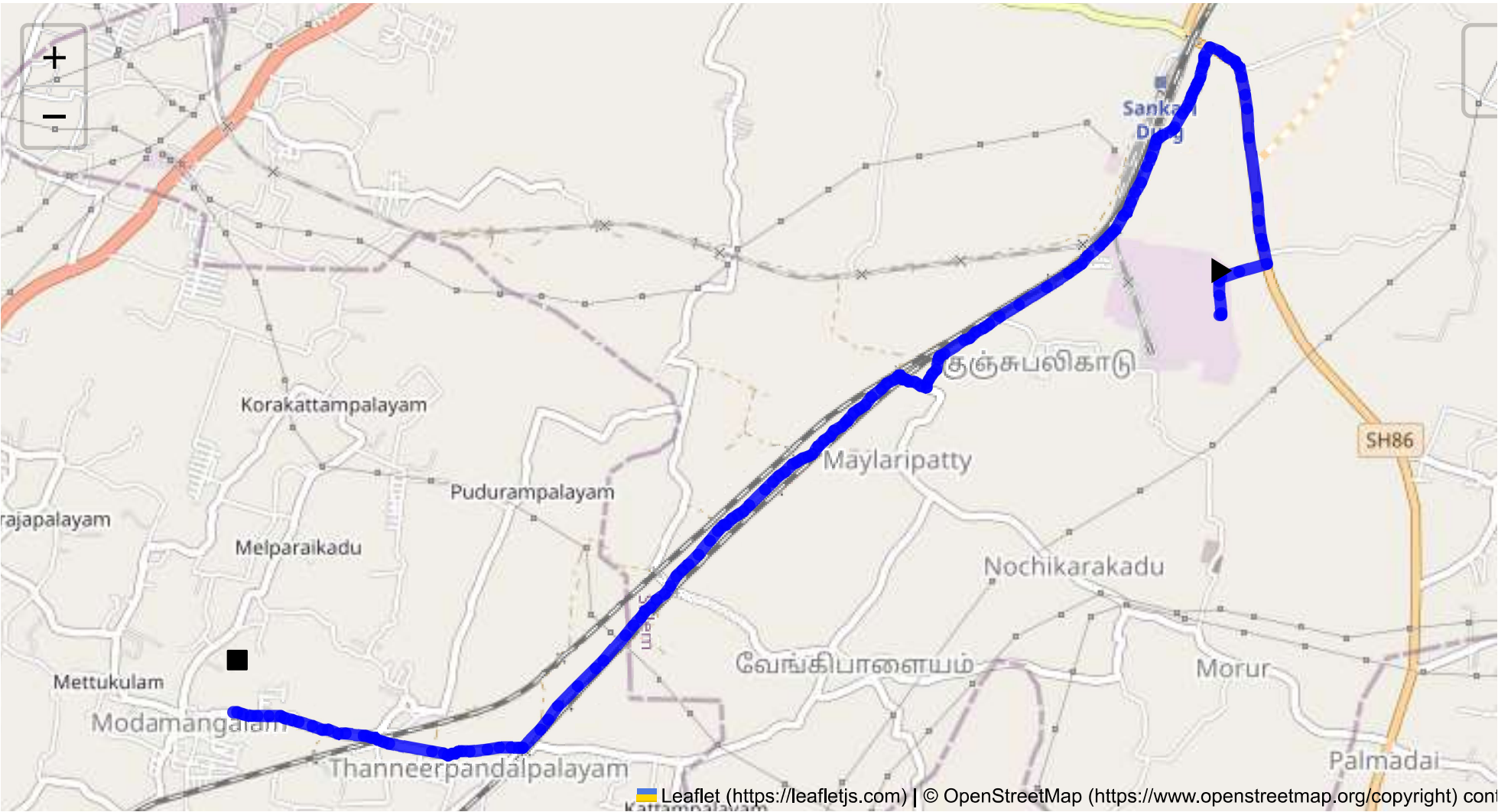
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 7.81 km
Estimated Duration: 0.3 hours
Adjusted Duration (Heavy Vehicle): 0.4 hours
Start: (11.4381, 77.8734)
End: (11.4214601, 77.8304734)



Welcome to the Journey Risk Management Study

Route Safety Analysis

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to CRCJ+F7H, Modamangalam Agraharam spans approximately 7.81 kilometers and involves traveling through rural and semi-urban areas. The main roads are likely local

roads with connections to state or district highways.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu typically has a tropical climate, experiencing hot and humid conditions most of the year. The monsoon season (October to December) can bring heavy rains, leading to potential flooding or water-logging on roads. High temperatures in summer months can also affect road conditions.

3. Analysis of Traffic Patterns

The route generally has moderate traffic but can experience congestion during local peak hours, mainly during early mornings and late evenings due to school and work commutes. Market days in nearby towns can further increase traffic volume.

4. Assessment of Road Quality and Infrastructure

The road quality is generally moderate, with some patches potentially needing repairs. There might be narrow stretches or sharp turns requiring careful navigation. Signage and road markings could be inconsistent, and street lighting may be limited in rural areas.

5. Suggestions for Alternative Routes for Emergencies

- Using parallel roads that connect through smaller villages may serve as alternative routes. However, these are typically narrow and may not be suitable for heavy vehicles.
- Main roads connecting to nearby towns like Tiruchengode or Erode may offer more secure detours depending on the nature of the emergency.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Regulations typically include restrictions on travel through densely populated areas during night hours and limits on certain routes for specific types of hazardous materials. Drivers must comply with safety guidelines, including proper vehicle signage and containment measures.

7. Overview of Historical Incidents

While specific historical data for this exact route is limited, rural and semi-urban areas in Tamil Nadu have seen incidents involving heavy vehicles due to poor visibility or road conditions. Incidents related to hazardous material remain rare but have notable consequences when they occur.

8. Environmental Considerations and Sensitive Areas

The region may have agricultural fields and villages sensitive to chemical spills or accidents involving hazardous materials. Care should be taken to prevent contamination and disruption to local ecosystems.

9. Analysis of Communication Coverage

Mobile network coverage is reasonably strong in most parts of Tamil Nadu, but there may be sporadic dead zones, especially in rural areas. Ensuring reliable communication channels for the entire journey is crucial for emergency preparedness.

10. Estimated Emergency Response Times

Emergency response times can vary:

- Urban and semi-urban areas: 15-30 minutes
- Rural stretches: 30-60 minutes or more due to longer distances and the availability of resources.

12. Overall Summary of Risk Assessment

This route poses a moderate risk level for transport of hazardous materials. Key risks include potential weather impacts, inconsistent road quality, and limited emergency support in rural areas. While traffic is generally not heavily congested, particular attention is needed during monsoon season and peak local travel hours. Effective communication, adherence to local regulations, and preparation for alternative routing are essential to lowering risks associated with this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Blind Spot	Blind Spot	11.44958, 77.87296	10 KM/Hr
5	Turn	Medium	11.44561, 77.87070	30 KM/Hr
6	Blind Spot	Blind Spot	11.43502, 77.86063	10 KM/Hr
7	Turn	Medium	11.43535, 77.85967	30 KM/Hr
8	Turn	Medium	11.43548, 77.85957	30 KM/Hr
9	Turn	Medium	11.43551, 77.85951	30 KM/Hr
10	Turn	High	11.41960, 77.84301	15 KM/Hr

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44958, 77.87296



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44561, 77.87070



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.43502, 77.86063



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43535, 77.85967



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43548, 77.85957



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43551, 77.85951



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41960, 77.84301